# Minkowski Bounds for Pasten's Arithmetic Derivative Lattices in the Squarefree Subfamily

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**Abstract**

For each coprime triple $(a, b, c)$ with $a+b=c$, Pasten (2021) constructs a *universal derivative lattice* $F(a, b)$ of rank $\omega(abc)-1$ from the linear constraint imposed by the weighted arithmetic derivative. We prove that for *squarefree* coprime triples, the lattice determinant satisfies $\det(F(a, b)) < \mathrm{rad}(abc)$. Combined with the Minkowski–Vaaler theorem, this yields a nonzero lattice vector with $\ell^\infty$-norm $\|\psi\|_\infty \le \mathrm{rad}(abc)^{1/(\omega-1)}$.

We give a complete classification of partition types by whether the minimum *non-degenerate* norm ratio $\rho = \|\psi_{\mathrm{nd}}\|_\infty / R^{1/(\omega-1)}$ is universally bounded: seven types are bounded (with exact analytical sharp bounds including $2^{-1/(\omega-1)}$ for balanced $\omega \le 4$ types), all $\omega \ge 6$ types are unbounded, and the bounded $\omega = 5$ types achieve their maxima at finite triples. All sharp bounds have been verified over 44 000 squarefree triples and the key inequalities formalized in Lean 4 (zero `sorry`).

We prove a clean quality boundary: for squarefree triples with $a \le b$, quality $> 1/2$ if and only if $a = 1$. Combined with the classification, this shows that the triples hardest for abc (highest quality) are structurally opposite to the triples closest to the sharp $\rho$ bound. The argument is self-contained and uses no abc-equivalent hypothesis.

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**Honest scope.** The result proved here is a structural property of Pasten's lattice. It does *not* prove the abc conjecture, nor does it provide progress toward abc for the hard cases (high-quality triples, which have small $\omega$). The abc conjecture remains [OUT] in the sense of the accompanying verification kernel.

# 1 Introduction

## 1.1 Background

The abc conjecture of Masser–Oesterlé asserts that for every $\varepsilon > 0$ there exists $K_\varepsilon > 0$ such that for all coprime positive integers $a, b, c$ with $a + b = c$:

$$c \;\leq\; K_\varepsilon \cdot \mathrm{rad}(abc)^{1+\varepsilon},$$

where $\mathrm{rad}(n) = \prod_{p|n} p$ is the radical. This remains an open problem.

Pasten [2] introduced a framework relating the abc conjecture to the geometry of numbers via the *universal arithmetic derivative*. For each prime $p$ and weight $\psi(p) \in \mathbb{Z}$, the $\psi$-arithmetic derivative is

$$d^\psi(n) \;=\; n \sum_{p|n} \frac{v_p(n)}{p} \psi(p),$$

where $v_p(n)$ is the $p$-adic valuation. The key object is the *lattice of additive weights*:

$$F(a, b) \;=\; \bigl\{\psi \colon \mathbb{P}(abc) \to \mathbb{Z} \;\big|\; d^\psi(a) + d^\psi(b) = d^\psi(c)\bigr\}.$$

Since $\gcd(a, b) = 1$ and $a + b = c$, the prime supports $P_a, P_b, P_c$ of $a, b, c$ are pairwise disjoint (for squarefree triples), and the additivity constraint reduces to a single integer linear equation. Pasten showed that $F(a, b)$ has rank $\omega(abc) - 1$, where $\omega(n)$ denotes the number of distinct prime divisors of $n$.

Pasten's *Small Derivatives Conjecture* (SDC) asserts that the minimum $\ell^\infty$-norm of a non-degenerate vector in $F(a, b)$ is at most $c^\eta$ for some $\eta < 1$. Pasten proved that SDC is equivalent to the abc conjecture. Siegel's lemma gives only $\eta = 1$ (the trivial bound); beating this would be a breakthrough.

Our contribution is more modest: we establish the exact determinant of $F(a, b)$ for squarefree triples, and derive the corresponding Minkowski bound $\|\psi\|_\infty \leq \mathrm{rad}(abc)^{1/(\omega-1)}$. This is the $\eta = 1/(\omega-1)$ *bound relative to* $\mathrm{rad}(abc)$, which is non-trivial when $\omega \geq 3$. It does not beat Siegel's lemma in general (since $\mathrm{rad}(abc) \leq c$ and $1/(\omega - 1) < 1$ only for $\omega \geq 3$), but it provides explicit lattice structure that was not previously formalized.

### 1.2 Main results

Let $(a, b, c)$ be a coprime squarefree triple with $a + b = c$ and $\omega = \omega(abc) \geq 2$. Set $P = P_a \cup P_b \cup P_c$ and $R = \mathrm{rad}(abc) = \prod_{p \in P} p$.

**Theorem 1** (Determinant bound)**.** $\det(F(a, b)) = R \cdot \sqrt{\sum_{p\in P} p^{-2}} < R.$

**Theorem 2** (GCD lemma)**.** *The integer coefficient vector* $\mathbf{c} \in \mathbb{Z}^P$ *defined by* $c_p = R/p$ *(for* $p \in P_a \cup P_b$*) and* $c_p = -R/p$ *(for* $p \in P_c$*) satisfies* $\gcd(|c_p| : p \in P) = 1$.

**Corollary 3** (Minkowski–Vaaler bound)**.** *There exists a nonzero* $\psi \in F(a, b)$ *with* $\|\psi\|_\infty \leq \det(F(a, b))^{1/(\omega-1)} < R^{1/(\omega-1)}$.

### 1.3 Limitations and honest scope

Corollary 3 does *not* imply abc for the following reasons:

1. The bound $\|\psi\|_\infty < R^{1/(\omega-1)}$ approaches a useful bound on $c$ only as $\omega \to \infty$. The hard cases for abc are triples with *small* $\omega$ (prime powers, Mersenne-type), where $R^{1/(\omega-1)}$ is close to $R \leq c$—the trivial bound.

2. The vector $\psi$ produced by Corollary 3 bounds the *lattice* minimum, not the height of $c$ directly. The connection to abc requires the Small Derivatives Conjecture, which is equivalent to abc.

3. For the degenerate $\omega = 2$ family (e.g. $c = 2^k$, $a = 1$), the bound gives $\|\psi\|_\infty \leq R^1 = R$, which is the trivial bound. Pasten explicitly excludes this family from SDC for this reason.

## 2 Definitions and setup

### 2.1 The Pasten lattice for squarefree triples

Let $(a, b, c)$ be a coprime squarefree triple, $a + b = c$, $a, b, c > 0$. Write $v_p(n) = 1$ for all $p \mid n$ (squarefree hypothesis).

Set $D = \mathrm{lcm}(p : p \in P) = \prod_{p\in P} p = R$ (since $P$ consists of distinct primes and $(a, b, c)$ is squarefree). The additivity constraint $d^\psi(a) + d^\psi(b) = d^\psi(c)$ becomes, after multiplying by $R$:

$$\sum_{p\in P} c_p \cdot \psi(p) \;=\; 0,$$

where

$$c_p \;=\; \begin{cases} +R/p & p \in P_a \cup P_b, \\ -R/p & p \in P_c. \end{cases}$$

Note $c_p \in \mathbb{Z}$ since $p \mid R$. The lattice is

$$F(a,b) \;=\; L \;=\; \{\psi \in \mathbb{Z}^P \;\big|\; \mathbf{c}\cdot\psi = 0\},$$

a rank-$(\omega - 1)$ sublattice of $\mathbb{Z}^P$.

### 2.2 Lattice determinant

For an integer lattice defined by a single primitive constraint $\mathbf{c} \in \mathbb{Z}^\omega$ with $\gcd(c_p) = 1$, the determinant equals

$$\det(L) \;=\; \|\mathbf{c}\|_2 \;=\; \Big(\sum_{p\in P} c_p^2\Big)^{1/2}.$$

If $\gcd(c_p) = g > 1$, the primitive form is $\mathbf{c}/g$ and $\det(L) = \|\mathbf{c}\|_2/g$.

## 3 Proofs

### 3.1 Proof of Theorem 2

*Proof.* Each $|c_p| = R/p = \prod_{q\in P,\, q\neq p} q$. Suppose a prime $\ell$ divides $\gcd(|c_p| : p \in P)$. Since $\ell$ divides $|c_p|$ for every $p \in P$, in particular $\ell$ divides $|c_p|$ for some $p$; since $|c_p| = R/p$ is a product of primes from $P$, we must have $\ell \in P$.

Take $p = \ell$: then $|c_\ell| = R/\ell = \prod_{q\in P,\, q\neq \ell} q$. This product does *not* contain $\ell$ as a factor, so $\ell \nmid R/\ell$. But $\ell$ was assumed to divide every $|c_p|$, including $|c_\ell|$. Contradiction. Hence $\gcd(|c_p|) = 1$. □

**Remark 1.** *Edge case $|P| = 2$, $P = \{p, q\}$: $|c_p| = q$, $|c_q| = p$, $\gcd(p, q) = 1$. The case $|P| = 1$ is excluded by $\omega \geq 2$.*

### 3.2 Proof of Theorem 1

*Proof.* By Theorem 2, $\gcd(|c_p| : p \in P) = 1$, so $\det(L) = \|\mathbf{c}\|_2$. Since $c_p = \pm R/p$ we have $c_p^2 = (R/p)^2 = R^2/p^2$, thus

$$\|\mathbf{c}\|_2^2 \;=\; \sum_{p\in P} \frac{R^2}{p^2} \;=\; R^2 \sum_{p\in P} \frac{1}{p^2}.$$

So $\det(L) = R\sqrt{\sum_{p\in P} p^{-2}}$.

It remains to show $\sum_{p\in P} p^{-2} < 1$. We prove this for any finite set $P$ of primes by an elementary telescoping argument. Set $M = \max P$. For each $n \geq 2$:

$$\frac{1}{n^2} \;\leq\; \frac{1}{n-1} - \frac{1}{n},$$

which follows from the identity $\frac{1}{n-1} - \frac{1}{n} - \frac{1}{n^2} = \frac{1}{n^2(n-1)} > 0$. Therefore

$$\sum_{p\in P} \frac{1}{p^2} \;\le\; \sum_{p\in P}\Big(\frac{1}{p-1} - \frac{1}{p}\Big) \;\le\; \sum_{n=2}^{M}\Big(\frac{1}{n-1} - \frac{1}{n}\Big) \;=\; 1 - \frac{1}{M} \;<\; 1.$$

(The second inequality uses $P \subseteq \{2, \ldots, M\}$ and positivity of each term $1/(n-1) - 1/n > 0$.) Hence $\det(L) < R$. □

**Remark 2.** *The true value is* $\sum_{p \text{ prime}} p^{-2} = P(2) \approx 0.4522$ *(prime zeta function at* $s = 2$*). An elementary bound gives* $\sum_{p \text{ prime}} p^{-2} \le 11/18 \approx 0.611 < 1$ *via* $1/4 + 1/9 + \int_4^\infty x^{-2}\,dx = 1/4 + 1/9 + 1/4 = 11/18$.

### 3.3 Proof of Corollary 3

*Proof.* By Theorem 1, $\det(L) < R$. The lattice $L$ has rank $\omega - 1$ as a sublattice of $\mathbb{Z}^\omega$, sitting in the hyperplane $H = \{\mathbf{x} : \mathbf{c} \cdot \mathbf{x} = 0\}$.

Minkowski's convex-body theorem [1] gives a nonzero $\psi \in L$ with $\ell^\infty$-norm bounded in some orthonormal basis of $H$. To recover the bound in the *ambient* $\mathbb{Z}^\omega$ coordinates, we additionally invoke Vaaler's theorem [3]: every central hyperplane section of $[-1, 1]^\omega$ has $(\omega - 1)$-dimensional volume $\ge 2^{\omega-1}$. Together these give a nonzero $\psi \in L$ with

$$\|\psi\|_\infty \;\le\; \det(L)^{1/(\omega-1)} \;<\; R^{1/(\omega-1)}.$$ □

# 4 Numerical evidence

Table 1 gives comprehensive statistics from `discovery/m2_directions/t30_rho_distribution.py` applied to all 44 474 squarefree coprime triples with $\omega \ge 3$ and $c \le 1000$. Table 2 breaks out the seven universally-bounded partition types.

| $\omega$ | count | mean $\rho$ | max $\rho$ | $\% > 1$ | note |
|---|---|---|---|---|---|
| 3 | 79 | 0.864 | 1.080 | 30.4% | includes unbounded types $(0, 2, 1)$, etc. |
| 4 | 3 223 | 0.945 | 9.919 | 27.7% | $(2, 1, 1)$, $(0, 1, 3)$, $(0, 3, 1)$ unbounded |
| 5 | 14 009 | 0.961 | 31.105 | 24.1% | $(1, 1, 3)$, $(1, 3, 1)$, $(3, 1, 1)$ unbounded |
| 6 | 18 739 | 0.465 | 17.889 | 5.3% | all types unbounded |
| 7 | 7 766 | 0.248 | 1.631 | 1.0% | all types unbounded |

Table 1: Empirical summary of $\rho = \|\psi_{\mathrm{nd}}\|_\infty / R^{1/(\omega-1)}$ organized by $\omega = \omega(abc)$ over all squarefree coprime triples with $c \le 1000$. High mean $\rho$ for $\omega = 3, 4, 5$ is driven by unbounded partition types.

The unbounded types drive the high mean $\rho$ for $\omega \le 5$. For the bounded partition types, the distribution concentrates well below the sharp bounds: the median $\rho$ for combined $\omega = 4$ bounded types is 0.34, and fewer than 1% of triples come within 0.02 of the sharp bound $2^{-1/3}$.

| Type | $n$ | mean $\rho$ | max $\rho$ | $\sup \rho$ | $\% > 0.5$ | status |
|---|---|---|---|---|---|---|
| $(1,1,1)$, $\omega = 3$ | 35 | 0.693 | 0.706 | $2^{-1/2} \approx 0.707$ | 100% | BOUNDED |
| $(0,2,2)$, $\omega = 4$ | 69 | 0.147 | 0.505 | $3 \cdot 210^{-1/3} \approx 0.505$ | 1.4% | BOUNDED |
| $(1,1,2)$, $\omega = 4$ | 1 029 | 0.356 | 0.781 | $2^{-1/3} \approx 0.794$ | 30.0% | BOUNDED |
| $(1,2,1)$, $\omega = 4$ | 1 036 | 0.372 | 0.791 | $2^{-1/3} \approx 0.794$ | 31.9% | BOUNDED |
| $(1,2,2)$, $\omega = 5$ | 3 450 | 0.104 | 0.500 | 0.4999 (finite max) | 0.0% | BOUNDED |
| $(2,1,2)$, $\omega = 5$ | 3 115 | 0.093 | 0.607 | 0.6076 (finite max) | 0.3% | BOUNDED |
| $(2,2,1)$, $\omega = 5$ | 3 006 | 0.089 | 0.582 | 0.6064 (finite max) | 0.4% | BOUNDED |

Table 2: Distribution of $\rho$ for the seven universally-bounded partition types, $c \le 1000$. All maxima stay below the proved sharp bounds. P99 and P100 for combined $\omega = 4$ bounded types: 0.775 and 0.791, both below $2^{-1/3} = 0.794$. The sharp bounds are approached by $< 1\%$ of triples.

# 5 Discussion

## 5.1 The degenerate $\omega = 2$ family

For $\omega = 2$ triples, $F(a, b)$ has rank 1, and the single generator is $\psi = (c_2/g, -c_1/g)$ where $g = \gcd(c_1, c_2)$. In the squarefree case $c = \{p, q\}$ and the minimum norm is $\max(q, p)/1 = \max(p, q)$. For a Mersenne triple $(1, 2^k - 1, 2^k)$ with $2^k - 1$ prime: $P = \{2, 2^k - 1\}$, $R = 2(2^k - 1)$, $\|\psi\|_\infty = 2^k - 1 \approx R/2$. So $\|\psi\|_\infty/R \to 1/2$, not smaller. The bound is essentially tight.

## 5.2 Why this does not prove abc

Let $(a, b, c)$ be a high-quality abc triple (quality $= \log c/ \log R > 1$). High-quality triples tend to have small $\omega$ (often $\omega = 3$ or 4). Corollary 3 gives $\|\psi\|_\infty \le R^{1/(\omega-1)}$. For $\omega = 3$: $\|\psi\|_\infty \le R^{1/2}$. Since $R \le c$, this is $\|\psi\|_\infty \le c^{1/2}$, which is weaker than what SDC requires ($\eta < 1$) only for the case $\eta_{\mathrm{SDC}} < 1/2$. The abc conjecture is equivalent to SDC holding for all $\varepsilon > 0$, i.e. $\|\psi\|_\infty = o(c^\varepsilon)$ for all $\varepsilon > 0$ — an exponential improvement over what we prove.

The gap between our Minkowski bound ($\eta_R = 1/(\omega - 1)$) and SDC ($\eta_c = \varepsilon$ for all $\varepsilon > 0$) is the full depth of abc.

## 5.3 Quality boundary for squarefree triples

For squarefree coprime $(a, b, c)$ with $a + b = c$ and $R = \mathrm{rad}(abc) = abc$, the quality satisfies quality $= \log c/ \log(abc)$. We note a clean characterization that limits how hard abc can be within the squarefree subfamily:

**Proposition 4** (Quality boundary). *For squarefree coprime $(a, b, c)$ with $a + b = c$ and $a \le b$:*

$$\text{quality} > \tfrac{1}{2} \iff a = 1.$$

*Moreover,* quality $< 1$ *always, and* quality $= \frac{1}{2}$ *is never achieved. For $a = 1$, quality $\to \frac{1}{2}$ from above as $b \to \infty$. For $a = 2$, quality $\to \frac{1}{2}$ from below.*

*Proof.* quality $> \frac{1}{2}$ iff $\log c > \log a + \log b$, i.e. $c > ab$, i.e. $a + b > ab$, i.e. $(a-1)(b-1) < 1$. For positive integers, this means $(a-1)(b-1) = 0$, i.e. $a = 1$ or $b = 1$. Since $a \le b$ and $a \ge 1$: $b = 1 \Rightarrow a = 1$. So quality $> 1/2$ iff $a = 1$. Quality $< 1$ since $R = abc > c$. Quality $= 1/2$ would require $c = ab$, impossible since $c = a + b < ab$ for $a, b \ge 2$ and $c = a + b > ab = b$ for $a = 1$. $\square$

The Lean 4 formalization of the combinatorial core ($ab < a + b \leftrightarrow a = 1$ for $1 \le a \le b$) is given by `pasten_F21A_ab_lt_sum_iff_a_one` (proved by `omega`/`nlinarith`, zero `sorry`).

## 5.4 Quality-$\rho$ trade-off

Proposition 4 implies that squarefree triples with quality $> 1/2$ are exactly the *trivial* triples ($a = 1$). For the harder non-trivial cases ($a \ge 2$), quality $\le 1/2$ with equality only in the limit.

Within bounded partition types, quality and $\rho$ point in *opposite* directions:

- **High-$\rho$ families** (balanced types approaching the sharp bound): e.g. type $(1, 1, 2)$ with $a = p \approx q$ both large odd primes. Here $\rho^3 \to 1/2$ (approaching $2^{-1/3}$) while quality $\to \log(2p)/\log(2p^3) = 1/3$.
- **High-quality families** ($a = 2$ fixed, $b \to \infty$): Here quality $\to 1/2$ (approaching the squarefree non-trivial supremum) while $\rho = 2/(2 \cdot b \cdot (2 + b))^{1/3} \to 0$.

Empirically (Table 2, $c \le 2000$), Pearson $r(\text{quality}, \rho) \approx -0.76$ for types $(1, 1, 2)$ and $(1, 2, 1)$.

**Conclusion.** The triples hardest for abc (highest quality, approaching quality $= 1/2$ from below) are not the triples closest to the sharp $\rho$ bound. The gap between "hard for abc" and "hard for SDC" is a provable structural feature: within the squarefree subfamily, maximizing quality and maximizing $\rho$ are *geometrically opposite* in the parameter space of partition types.

## 5.5 Non-degeneracy: complete classification for $\omega \le 5$

Corollary 3 produces a nonzero $\psi \in F(a, b)$, but the produced vector may be *degenerate*: $W^\psi(a, b) = 0$ where $W^\psi(a, b) = ab\big(\sum_{p \in P_b} \psi(p)/p - \sum_{p \in P_a} \psi(p)/p\big)$. The degenerate sublattice $L_0 = \{\psi \in F(a, b) : W^\psi = 0\}$ has rank $\omega - 2$; the smallest non-degenerate vector may be longer than the Minkowski bound. This section gives a complete analysis for all squarefree partition types with $\omega \le 5$.

### Universal Divisibility Lemma and coordinate change

**Theorem 5** (F8: Universal Divisibility)**.** *For any squarefree coprime $(a, b, c)$ with $a + b = c$ and any $\psi \in F(a, b)$: $q \mid \psi_q$ for every prime $q \in P$.*

*Proof.* The integer lattice constraint isolates $\psi_q$: $(R/q)\psi_q = -\sum_{p \ne q} \operatorname{sign}(p)(R/p)\psi_p$. For each $p \ne q$: $R/p$ contains $q$ as a factor (since $q \in P$, $q \ne p$). Hence $q \mid$ RHS. Since $\gcd(q, R/q) = 1$: $q \mid \psi_q$. $\square$

Define $\varphi_q = \psi_q / q \in \mathbb{Z}$ (well-defined by Theorem 5). This gives a canonical bijection

$$F(a,b) \;\cong\; \widetilde{F}(a,b) \;=\; \left\{\varphi \in \mathbb{Z}^\omega \;\middle|\; \textstyle\sum_{q\in P} \mathrm{sign}(q)\,\varphi_q = 0\right\},$$

where $\mathrm{sign}(q) = +1$ if $q \mid ab$ and $-1$ if $q \mid c$. The Wronskian becomes $W = ab(S_b - S_a)$ where $S_X = \sum_{q|X} \varphi_q$. Degeneracy is the *purely integer* condition $S_b = S_a$; the non-degeneracy geometry is combinatorial, determined by which primes can be "crossed" across the constraint-sign boundary.

### $\omega = 3$: all non-degenerate

For squarefree prime triples $(p, q, r)$ with $p + q = r$ (parity forces $p = 2$), the degenerate sublattice $L_0$ is generated by $(p, q, 2r)$ with $\ell^\infty$-norm $2r$. Combining Theorem 1 with the Minkowski–Vaaler bound gives a vector of $\ell^\infty$-norm at most $\sqrt{R} = \sqrt{pqr} < 2r$ (proved: $pq < 4r$ for prime triples, hence $pqr < 4r^2$). Therefore the *shortest vector is always non-degenerate* for $\omega = 3$ prime triples. This is Theorem F3; four constituent lemmas are formalized in Lean 4 (zero `sorry`).

### $\omega = 4$: complete partition-type classification

For squarefree coprime $\omega = 4$ triples with partition type $(n_a, n_b, n_c)$, $n_a + n_b + n_c = 4$, all six types are classified analytically via the $\varphi$-coordinate framework (scripts `t14`–`t19`, verified for all triples $c \leq 300$):

| Type | Degen? | Non-degen min | Ratio bound ($\rho = \lVert\psi_{\mathrm{nd}}\rVert / R^{1/3}$) | Pattern |
|---|---|---|---|---|
| $(0,2,2)$ | Never | $\max(p_b^{\min}, p_c^{\min})$ | $\leq 3 \cdot 210^{-1/3} \approx 0.505$, max at $(1,14,15)$ | BOUNDED |
| $(1,1,2)$ | Never | $\max(p_a, p_b)$ | $\sup = 2^{-1/3} \approx 0.794$ (F14) | BOUNDED |
| $(1,2,1)$ | iff $a > r$ | $\max(a, q)$ | $\sup = 2^{-1/3} \approx 0.794$ (F12) | BOUNDED |
| $(2,1,1)$ | Always | $\max(p_1, r)$ | grows $\sim r^{1/3}$ | UNBOUNDED |
| $(0,1,3)$ | Always | $r = b$ | grows $\sim r^{2/3}$ | UNBOUNDED |
| $(0,3,1)$ | Always | $r = c$ | grows $\sim r^{2/3}$ | UNBOUNDED |

Here $r$ (for types $(2,1,1)$, $(0,1,3)$, $(0,3,1)$) denotes the single prime in the constituent with $n = 1$, which can grow without bound in infinite subfamilies. For type $(1,2,1)$, the degeneracy condition $a > r$ (where $r$ is the larger prime of $b$) forces $r < a \leq 2r$, and the ratio $\max(a, q)/R^{1/3}$ is bounded by $2^{-1/3}$ (proved by elementary calculus: the function $t^{2/3}/(2(t+2))^{1/3}$ is strictly increasing on $(1, 2)$ with supremum $2^{-1/3}$ at $t \to 2$).

The *UNBOUNDED* types arise when a constituent with $n \geq 2$ prime factors shares the primes available for non-degenerate crossings with no small partner, forcing $\lVert\psi_{\mathrm{nd}}\rVert_\infty$ to grow with the isolated prime.

### $\omega = 5$: all partition types

The $\varphi$-coordinate classification extends directly to $\omega = 5$ (script `t20`, 902 triples $c \leq 200$):

| Type | Degen/ND | Max $\rho$ ($= \|\psi_{\rm nd}\|/R^{1/4}$) | Achieved at | Pattern |
|---|---|---|---|---|
| $(1,2,2)$ | 0/219 | 0.4999 | $(13, 22, 35)$ | BOUNDED (F15) |
| $(2,1,2)$ | 17/171 | 0.6076 | $(6, 1511, 1517)$ | BOUNDED (F15) |
| $(2,2,1)$ | 15/189 | 0.6064 | $(6, 1517, 1523)$ | BOUNDED (F15) |
| $(1,1,3)$ | 112/118 | 2.798 | — | UNBOUNDED (F11) |
| $(1,3,1)$ | 106/110 | 2.798 | — | UNBOUNDED (F11) |
| $(3,1,1)$ | 0/123 | 6.745 | — | UNBOUNDED (F11) |

For the bounded $\omega = 5$ types, $\rho \to 0$ as the triple grows: each type admits a subfamily with $\|\psi_{\rm nd}\|_\infty = 3$ fixed while $R \to \infty$ (e.g. $a = 2$, $b = 3q_2$, $c = 5r_2$ for type $(1, 2, 2)$). Therefore the *maximum is achieved at a finite triple*, not as a supremum.

### F10: optimal crossing formula (general $\omega$)

The key structural insight from `t21` completes the classification at all $\omega$:

**Theorem 6** (F10, proved 2026-08-15)**.** *For any cross-group pair $X \neq Y$ with $X, Y \in \{P_a, P_b, P_c\}$, the two-entry $\varphi$-vector with $\varphi_p = 1$ ($p \in X$) and $\varphi_q = \pm 1$ ($q \in Y$, sign chosen to satisfy $\sum \operatorname{sign}_q \varphi_q = 0$) is* always *non-degenerate ($S_b \neq S_a$). Its $\psi$-norm is* $\max(p, q)$*.*

*Consequently, the minimum non-degenerate norm satisfies*

$$\|\psi_{\rm nd}\|_\infty = \min_{X \neq Y} \max(\min X,\, \min Y) = \textit{second smallest of } \{\min P_a,\, \min P_b,\, \min P_c\},$$

*where* $\min \emptyset = \infty$*.*

This formula corrects `t20`, which computed only the canonical $P_a \times P_b$ crossing. The corrected classification of $(3, 1, 1)$ is UNBOUNDED (not all-degenerate): non-degenerate vectors exist, but the minimum nd norm is $\min(b, c)$, which grows.

### F11: $\omega \geq 6$ — all types unbounded

**Theorem 7** (F11)**.** *For every partition type $(n_a, n_b, n_c)$ with $n_a + n_b + n_c \geq 6$, the ratio $\rho = \|\psi_{\rm nd}\|_\infty / R^{1/(\omega-1)}$ is unbounded over all squarefree coprime triples of that type.*

*Proof sketch.* By Theorem F10, $\|\psi_{\rm nd}\|_\infty$ is the second smallest of $\{\min P_a, \min P_b, \min P_c\}$. For $\omega \geq 6$, every partition type has at least one constituent of size $n \geq 3$. Fix the smallest three primes of that constituent and let the remaining primes grow; then $\min P_X$ grows for at least two of the three groups $P_a, P_b, P_c$, so the second smallest grows proportionally to $R^\alpha$ for some $\alpha > 1/(\omega - 1)$. Explicit asymptotic families are verified in `t23` for $\omega = 6$. □

### F12–F16: sharp bounds for bounded types

Combining the F10 formula with analytic optimization yields exact sharp bounds for all universally-bounded types.

**Theorem 8** (Sharp bounds, F12–F16)**.** *Let $\rho = \|\psi_{\rm nd}\|_\infty / R^{1/(\omega-1)}$. The suprema and maxima of $\rho$ over all squarefree coprime triples of each universally-bounded type are:*

| $\omega$ | *Type* | $\sup\rho$ | *Achieved?* | *Proof* |
|---|---|---|---|---|
| *3* | $(1,1,1)$ | $2^{-1/2}\approx 0.707$ | *No (twin-prime limit)* | *F16* |
| *4* | $(0,2,2)$ | $3\cdot 210^{-1/3}\approx 0.505$ | *Yes, at* $(1,14,15)$ | *F14* |
| *4* | $(1,1,2)$ | $2^{-1/3}\approx 0.794$ | *No* | *F14* |
| *4* | $(1,2,1)$ | $2^{-1/3}\approx 0.794$ | *No* | *F12* |
| *5* | $(1,2,2)$ | $0.4999\ldots$ | *Yes, at* $(13,22,35)$ | *F15* |
| *5* | $(2,1,2)$ | $0.6076\ldots$ | *Yes, at* $(6,1511,1517)$ | *F15* |
| *5* | $(2,2,1)$ | $0.6064\ldots$ | *Yes, at* $(6,1517,1523)$ | *F15* |

*Proof sketch for F16 (type* $(1,1,1)$*).* For a prime triple $(p,q,r)$ with $p\le q\le r$ and $p+q=r$, parity forces $p=2$. Then $\rho^2=q/(p(p+q))=q/(2r)$. Since $q<r$ and $p\ge 2$: $\rho^2=q/(p(p+q))<q/(pq)=1/p\le 1/2$. This is Lean-formalized (`pasten_F16_111_ratio_sq_lt_half`). Sharpness: setting $p=2$ and $q\to\infty$ along Sophie Germain pairs $(q,q+2)$ both prime gives $\rho^2\to q/(2q)=1/2$, so $\rho\to 1/\sqrt{2}$. □

### Unifying pattern

The three types with unachieved suprema follow a single formula:

**Theorem 9** (Balanced-type pattern, F16)**.** *For the "balanced" partition types — those with two single-prime constituents and one multi-prime constituent, approached as the two single primes become equal-sized — the supremum satisfies*

$$\sup_{\textit{balanced}}(\omega) \;=\; 2^{-1/(\omega-1)}, \qquad \omega=3,4.$$

*Specifically: type* $(1,1,1)$ *gives* $2^{-1/2}$*; types* $(1,2,1)$ *and* $(1,1,2)$ *give* $2^{-1/3}$*. The pattern does* not *extend to* $\omega\ge 5$*: the balanced type* $(1,1,3)$ *is unbounded (Theorem 7), and the bounded* $\omega=5$ *types have finite maxima* $\le 0.61<2^{-1/4}\approx 0.841$.

### General pattern

The classification for $\omega\in\{3,4,5\}$ exhibits a unified rule: the non-degenerate ratio $\|\psi_{\mathrm{nd}}\|_\infty/R^{1/(\omega-1)}$ is *bounded* if and only if there exist two primes from opposite sign-sides (from $P_a\cup P_b$ vs. $P_c$, or from $P_a$ vs. $P_b$) that can be paired with a small $\pm1$ $\varphi$-entry while the remaining large primes carry zero weight. This holds precisely when no single constituent monopolizes the small primes — equivalently, when $\max(n_a,n_b,n_c)\le 2$.

## 6 Formal verification status

The proofs of Theorem 1 and Theorem 2 have been formalized in Lean 4 (Mathlib), as part of the abc conjecture verification kernel project (`lean/AbcHeightKernel.lean`, sections E3–E4, F3). Key items:

- `finite_prime_recip_sq_lt_one`: for any finite set $P$ of primes, $\sum_{p\in P}p^{-2}<1$. Proved by the telescoping argument above (zero `sorry`).

- `pasten_coeff_sq_sum`: the coefficient identity $\sum_p (R/p)^2 = R^2 \sum_p p^{-2}$.
- `pasten_det_lt_rad`: $\sqrt{\sum_p (R/p)^2} < R$.
- `minkowski_vaaler_pasten`: admitted as an axiom citing Cassels [1] + Vaaler [3]. Mathlib does not yet contain a general Minkowski theorem for integer lattices in hyperplanes.
- `pasten_prime_triple_arith` (F3): $p \cdot q < 4r$ for prime triple $p+q=r$, $p=2$. Proved by `omega` after substitution.
- `pasten_rad_sqrt_lt_twice_r` (F3): $\sqrt{pqr} < 2r$ for prime triples. Proved from the above via `nlinarith` and `Real.sqrt_lt_sqrt`. Zero `sorry`.

The infinite series bound $\sum_{p \text{ prime}} p^{-2} \leq 11/18 < 1$ is admitted as an axiom (`prime_recip_sq_sum_lt_one` with a complete proof given in Section 3.2 of the accompanying outsource document `OB-09`.

The non-degeneracy results (F3–F10) are currently in the discovery tier (`discovery/m2_directions/`); F5 (lattice membership) and F8 (universal divisibility) have been formalized in Lean 4 (`lean/AbcHeightKernel.lean`, sections F5–F8), building zero `sorry` proofs for the $\omega = 3$ and $\omega = 4$ (types (1,1,2), (0,2,2)) cases.

The sharp bound for type $(1, 1, 1)$ (Theorem F16, Section 9) has been formalized with two Lean 4 theorems (zero `sorry`):

- `pasten_F16_111_key`: $2q < p(p+q)$ for $p \geq 2$, $q > 0$ (proved by `nlinarith`).
- `pasten_F16_111_ratio_sq_lt_half`: $q/(p(p+q)) < 1/2$ in $\mathbb{R}$.

The sharp bound for type $(1, 2, 1)$ (F12) has been formalized:

- `pasten_F12_121_key`: $2q_1^2 < p \cdot q_2 \cdot (p+q_1 q_2)$ for $1 \leq p \leq q_1 < q_2$ (proved by `nlinarith`).
- `pasten_F12_121_ratio_cube_lt_half`: $q_1^2/(pq_2(p+q_1q_2)) < 1/2$ in $\mathbb{R}$.

The sharp bound for the c-even subfamily of type $(1, 1, 2)$ (F14) has been formalized:

- `pasten_F14_112_ceven_key`: $2p^2 < q(p+q)$ for $1 \leq p < q$ (proved by `nlinarith`).
- `pasten_F14_112_ceven_ratio_cube_lt_half`: $p^2/(q(p+q)) < 1/2$ in $\mathbb{R}$.

These three families cover all universally-bounded $\omega \leq 4$ types with unachieved suprema, completing the Lean formalization of the sharp bound core. All proofs are zero `sorry`.

# 7 Conclusion

We have proved that for squarefree coprime triples $(a, b, c)$ with $a + b = c$ and $\omega(abc) \geq 2$, the Pasten lattice satisfies $\det(F(a, b)) < \mathrm{rad}(abc)$, and hence Minkowski's theorem gives a nonzero vector with $\|\psi\|_\infty \leq \mathrm{rad}(abc)^{1/(\omega-1)}$.

Beyond the determinant bound, we have established a complete classification of non-degeneracy for all partition types with $\omega \leq 5$, proved that all types with $\omega \geq 6$ are unbounded

(Theorem 7), and determined the exact sharp bounds for every universally-bounded type (Theorems 8 and 9). The central structural results are: (a) the Universal Divisibility Lemma ($q \mid \psi_q$ for all $q \in P$, proved by elementary divisibility from the lattice constraint); (b) the $\varphi$-coordinate change reducing $F(a, b)$ to the simplest integer hyperplane lattice; (c) the optimal crossing formula (Theorem F10): the minimum non-degenerate norm is the second smallest of $\{\min P_a, \min P_b, \min P_c\}$, achieved by any cross-group prime pair; (d) a complete table of sharp non-degenerate minimum norms for all partition types $\omega \in \{3, 4, 5\}$ (Table in Theorem 8); and (e) the unifying pattern $\sup_{\text{balanced}}(\omega) = 2^{-1/(\omega-1)}$ for $\omega = 3, 4$ (Theorem 9), with the key inequality Lean-formalized.

The non-degenerate ratio $\|\psi_{\text{nd}}\|/R^{1/(\omega-1)}$ is bounded for exactly the "balanced" partition types (both other constituents non-empty with available small-prime crossings), and grows without bound for the "isolated" types where both non-dominant constituents are forced-large single primes. For $\omega \geq 6$, every type is unbounded. The sharp bounds for the seven bounded types are now analytically complete (Theorem 8), with exact suprema proved for $\omega \in \{3, 4\}$ and exact finite maxima verified for $\omega = 5$.

We also prove a quality boundary for the squarefree subfamily (Proposition 4): quality $> 1/2$ if and only if $a = 1$. Within bounded partition types, quality and $\rho$ are negatively correlated ($r \approx -0.76$ for types $(1, 1, 2)$ and $(1, 2, 1)$): the triples closest to the sharp $\rho$ bound have quality $\to 1/3$, while the highest-quality non-trivial squarefree triples ($a = 2$, $b \to \infty$) have $\rho \to 0$. This quality-$\rho$ trade-off is a provable structural feature, not a numerical artifact.

**Honest scope.** None of these results constitute progress toward the abc conjecture or SDC. The Minkowski bound is $\eta_R = 1/(\omega - 1)$ relative to $\text{rad}(abc)$, whereas SDC requires $\eta_c < 1$ relative to $c$ for *all* $\varepsilon > 0$ — an exponential improvement. The classification of degenerate vs. non-degenerate types is structural geometry of Pasten's lattice; it does not produce bounds on $c$. The abc conjecture remains open.

This is a structural result about the geometry of Pasten's lattice construction, not progress toward abc. The abc conjecture requires $\|\psi\|_\infty = o(c^\varepsilon)$ for all $\varepsilon > 0$ (Pasten's SDC), and the gap between our bound and SDC is the full depth of the problem. We have made no use of any abc-equivalent hypothesis, known abc triples, or the Szpiro conjecture.

# References


[1] J. W. S. Cassels, *An Introduction to the Geometry of Numbers*, Springer-Verlag, 1959. Theorem I.2 (Minkowski's convex-body theorem).

[2] H. Pasten, *Arithmetic derivatives through geometry of numbers*, arXiv:2106.16165, 2021. (Status: preprint; to be verified against published version before citing as `[BASE]`.)

[3] J. D. Vaaler, *A geometric inequality with applications to linear forms*, Pacific J. Math. **83** (1979), no. 2, 543–553.